Neuromuscular Training Improves Biomechanical Deficits at the Knee in Anterior Cruciate Ligament–Reconstructed Athletes

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Abstract

Objective: Athletes who return to sport after anterior cruciate ligament reconstruction (ACLR) demonstrate persistent biomechanical and neuromuscular deficits of the knee. There is limited evidence on what effect a neuromuscular training (NMT) program has on knee biomechanics in a cohort of athletes with ACLR. Therefore, the primary aim of this study was to quantify the effect of an NMT program on knee biomechanics in a cohort of ACLR athletes. Second, the post-training knee biomechanics were compared between the cohort of ACLR and control athletes. **Design:** Cohort study. **Setting:** Controlled laboratory setting. **Participants:** Eighteen athletes with ACLR and 10 control athletes. **Interventions:** Neuromuscular training. **Main Outcome Measures:** Knee kinematics and kinetics during a double-limb jump-landing task. **Results:** There were no significant interactions ($P > 0.05$) observed for the athletes with ACLR. However, there was a significant main effect of biomechanics testing session ($P < 0.05$) for knee flexion angle and moments; athletes with ACLR demonstrated greater knee flexion angle and lower knee flexion moment during the post-training biomechanics testing session. Post-training comparison between the ACLR and control athletes demonstrated no significant interactions ($P > 0.05$) between the groups. There was a significant main effect of group ($P < 0.05$) for knee frontal angle, as athletes with ACLR landed with greater knee adduction than the control athletes. **Conclusions:** Significant improvements in knee sagittal plane biomechanical measures were observed after the NMT program by the athletes with ACLR. In addition, post-training comparison of the ACLR and control groups demonstrates comparable knee biomechanics.

Key Words: neuromuscular training, anterior cruciate ligament, anterior cruciate ligament reconstruction, biomechanics, drop vertical jump

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INTRODUCTION

Young athletes who return to sport (RTS) after anterior cruciate ligament reconstruction (ACLR) have a greater propensity for poor outcomes.[1] A systematic review on the current state of play after ACLR indicates that only 65% of ACLR athletes return to their preinjury level of activity and only 55% of these athletes return to competitive level of sports.[2] However, the athletes who return to high levels of sport are highly susceptible to a second injury.[3–5] The emerging evidence indicates that nearly 1 in 3 to 4 young, active athletes who return to their respective sport will sustain an ACL graft failure or injury to the contralateral ACL after.[6,7] These same ACLR athletes are 30 to 40 times more likely to suffer a second ACL injury than an uninjured counterpart is to sustain their first.[5] These consequences pale in comparison with the long-term joint morbidity that a majority of ACLR athletes are confronted with. The early, premature breakdown of the articular cartilage is reported in 50% to 90% of athletes in 10 to 20 years after ACLR.[8–10] The etiology of these poor outcomes is possibly related and may be due to the chronic, abnormal movement patterns found in ACLR athletes.[7,11,12]

Deficits in lower extremity biomechanics and neuromuscular control after ACLR are commonly reported in these athletes. A recent systematic review on knee movement patterns during gait found that joint kinematics are on average restored 6 years after ACLR and other sagittal plane kinetic measures remain lower than a control population, which indicates that normal knee function may not ever be restored.[13] Abnormal limb loading strategies are also observed in a range of functional activities including during squatting,[14,15] running,[16] and landing[7,17–20] movements. Notably, these residual movement deficits persist at the time when athletes return to activity and are also directly associated with an elevated risk of second ACL injury.[7,21–23] A prospective study that screened ACLR athletes just before the return of sport identified several hip and knee neuromuscular deficits, which went on to predict second ACL injuries with high sensitivity and specificity.[7] The pervasive functional deficits in this population indicate that rehabilitation of the athletes back to sport to preinjury level of health may not be

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sufficient. This highlights the importance of augmenting postoperative rehabilitation with training programs to address residual movement deficits before athletes returning to sport.

The current evidence indicates that neuromuscular training (NMT) programs can address functional deficits in athletes.[24] Previous studies that have implemented NMT in populations vulnerable to knee injuries have modified potential biomechanical risk factors, improved neuromuscular control, and decreased ACL injury incidence.[25–28] Furthermore, a systematic review and meta-analysis on the effectiveness of ACL injury prevention training programs found strong evidence in support of these programs.[26] The pooled estimates from this study found a risk reduction of 52% for female athletes and 85% for male athletes.[26] These same benefits from NMT may allow ACLR athletes to safely and effectively return to sport. Ideally, this training program would be implemented at the end stage of rehabilitation or just before returning athletes back to sport. However, it is first important to understand what effect an NMT program has on knee biomechanics in a group of young athletes with ACLR before returning to sport. The primary aim of this study was to quantify the effect of an NMT program on knee sagittal (knee flexion angle and moment) and frontal (knee abduction angle and moment) plane angle and moments in a cohort of ACLR athletes since these measures have been implicated in greater risk of ACL injury. Furthermore, we compared the same knee kinematic and kinetic measures between a group of ACLR and control athletes after both groups completed the training program. The hypothesis tested was that the ACLR group and their limbs (involved and uninvolved) would demonstrate greater knee flexion angles, lower frontal plane knee angles, and decreased knee flexion and abduction moments after participation in the NMT program. The second hypothesis tested was that the ACLR group and their limbs would demonstrate sagittal and frontal plane knee angles and moments that were not significantly different than a group of uninjured controls and their limbs (dominant and nondominant) after NMT.

METHODS

Subjects and Clinical Criteria to Enroll in Study

Eighteen (n = 18) ACLR athletes and 10 (n = 10) uninjured controls were enrolled in this study (Table 1). The study was approved by the university's Institutional Review Board, and written informed consent was received from all athletes. Informed assents and parental permission were provided for athletes who were younger than 18 years. All the enrolled ACL-injured athletes in the study received a hamstring tendon autograft during ACLR. Postoperative rehabilitation was completed at the university's affiliated Sports Medicine clinics. The athletes with ACLR were approximately 8 months (7.7 ± 3.7 months) out from surgery at the beginning of the study. Before beginning baseline biomechanics testing, both ACL-injured and ACL-uninjured athletes were subject to a bilateral clinical evaluation conducted by a licensed physical therapist or athletic trainer. Clinical examinations were performed by licensed physical therapists trained in the clinical examination protocol and criteria to perform the testing session. The purpose of the clinical examination is to ensure the health of the knee of each participant to safely participate in the high demand testing and training. The examination included measurements of active and passive knee joint range of motion, knee joint effusion, isokinetic knee extensor and flexor strength test (Biodex System 3; Biodex, Shirley, NY) at 60 degrees/sec, and 5 continuous bilateral single-leg hops for maximum vertical height. To participate in the study, each athlete had to demonstrate: (1) pain free knee range of motion, (2) trace or no knee joint effusion,[29] (3) <30% knee extensor strength deficit,[30] and (4) the willingness to single-leg hop in place without any pain or loss of balance. Athletes who demonstrated biomechanics excessively risky or unsafe (ie, large medial–lateral knee excursion or little to no knee flexion) during the single-leg hop in place were referred back to their physical therapist.

Biomechanical Testing and Neuromuscular Training

Biomechanical assessment was performed before enrollment in the NMT program and directly after completion. During this evaluation, athletes were fitted with 55 retroreflective markers and performed 3 successful drop vertical jumps (DVJs) off a 30.5-cm plyometric box onto embedded force plates (Bertec, Corp, Columbus, OH). Marker trajectories were sampled at 240 Hz by a 12 camera motion-capture system (Motional Analysis Corporation, Santa Rosa, CA), and separate ground reaction force data were collected for each limb at 1200 Hz.

Each athlete from the ACLR and control group completed the 12-session NMT program that focused on enhancing trunk stability, increasing and coordinating dynamic lower extremity joint flexion, and optimizing landing mechanics as described in detail by Di Stasi et al.[31] The program was administered by study personnel trained specifically in the implementation of the NMT protocol by a licensed physical therapist (S.D.). These study personnel included athletic trainers, strength and conditioning specialists, physical therapists, and graduate students in the laboratory. In summary, the NMT program included 7 separate progressions that involved both unilateral and bilateral lower-limb exercises and core control and strengthening work-outs (Table 2). In addition, the 7 progressions included 4 phases of increasing difficulty. Readiness to progress from one phase to the next was determined by the clinician or trainer on an exercise-by-exercise basis. Progression was based on the athlete's ability to demonstrate proper form for at least 80% of the total repetitions.

TABLE 1. Demographics of Study Cohorts

Group (Males/Females)	Age (yrs)	Weight (kg)	Height (m)
ACLR group (M: 8; F: 10)	19.4 ± 7.2	72.3 ± 15.4	1.68 ± 0.1
Control group (M: 4; F: 6)	16.0 ± 3.7	73.1 ± 24.4	1.66 ± 0.1

There were no significant differences between groups.

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TABLE 2. A Summary of the Neuromuscular Training Program

Exercise Progression	Tasks
Single-leg hop	Phase I: single-leg lateral hop hold on Airex mat
	Phase II: single-leg 90-degree hop hold on Airex mat
	Phase III: single-leg lateral BOSU (round) hop hold
	Phase IV: single-leg 90-degree Airex hop hold reaction ball catch
Single-leg anterior	Phase I: step hold
	Phase II: jump-single-leg hold
	Phase III: hop hold
	Phase IV: hop-hop-hold
Romanian deadlift	Phase I: single-leg deadlift
	Phase II: single-leg deadlift on Airex
	Phase III: single-leg deadlift on BOSU
	Phase IV: single-leg deadlift with dumbbells
Lunge	Phase I: walking lunges
	Phase II: backward lunge walk
	Phase III: lunge jumps
	Phase IV: scissor jumps
Double-leg jump	Phase I: box butt touch
	Phase II: squat jumps
	Phase III: single tuck jump with hold
	Phase IV: double tuck jumps with hold
Prone trunk stability	Phase I: BOSU (round) toe-touch swimmers
	Phase II: BOSU (round) with partner perturbation
	Phase III: prone bridge (elbows and knees) hip extension w/opposed shoulder flexion
	Phase IV: prone bridge (elbows and toes) hip extension
Lateral trunk flexion	Phase I: BOSU (round) lateral crunch
	Phase II: box lateral crunch
	Phase III: BOSU (round) lateral crunch with ball catch
	Phase IV: Swiss ball lateral crunch

Data Treatment and Statistical Analysis

Customized software was used to reduce and analyze kinematic and kinetic data. Marker position gaps that were within the 25 consecutive frames during the jump-landing task were filled using a cubic spline function in Cortex, a motion-capture software (Cortex version 4.1, Motion Analysis, Corp, Santa Rosa, CA). After all the markers were properly labeled and the gaps were filled, these data along with the ground reaction force data were exported to Visual 3D (C-motion, Inc, Germantown, MD) where customized static models scaled were first generated to each participant's anthropometric measurements. Subsequently, the marker position data and ground reaction force data were low-pass filtered using a bidirectional Butterworth filter at 12 Hz and 50 Hz, respectively. The hip joint centers were determined using a validated, anthropometric calculation,[32] and a virtual marker was created in postprocessing based on the relative location of the calculated hip joint center to other markers on the pelvis segment. All data were time-normalized to 100% of stance. Cardan–Euler sequence for local coordinated systems (X-Y-Z) was used to calculated kinematic variables, and inverse dynamics was used to calculate kinetic variables. Initial contact (IC) was defined when the vertical component of the ground reaction force exceeded 10 N. The kinematic and kinetic calculations were processed using custom codes in Visual 3D and Matlab (Mathworks, Inc, Natick, MA).

The analysis of knee kinematic and kinetic variables was focused on IC and peak. Initial contact is of interest because injuries have been demonstrated to occur within approximately 20 to 50 milliseconds of landing.[33] Repeated-measures analysis of variance (ANOVA) was performed to assess interactions and main effects of session (pre-training and post-training) and limb (involved/dominant and uninvolved/nondominant) to understand the effects of NMT in the ACLR cohort. A 2-way ANOVA was used to assess the interactions of group (ACLR vs control) and limb after the NMT intervention. Post hoc paired and independent *t* tests were used to test for significant differences between limb and session, and groups, respectively. Limb symmetry index [(LSI = involved/uninvolved) × 100] was calculated for sagittal plane kinematic and kinetic variables during IC and peak for the athletes with ACLR before and after training. Paired *t* tests were used to compare average LSI of athletes with ACLR before participating in the training and after completing the training program. The alpha level was set to 0.05 a priori to determine significant results (Figure 1).

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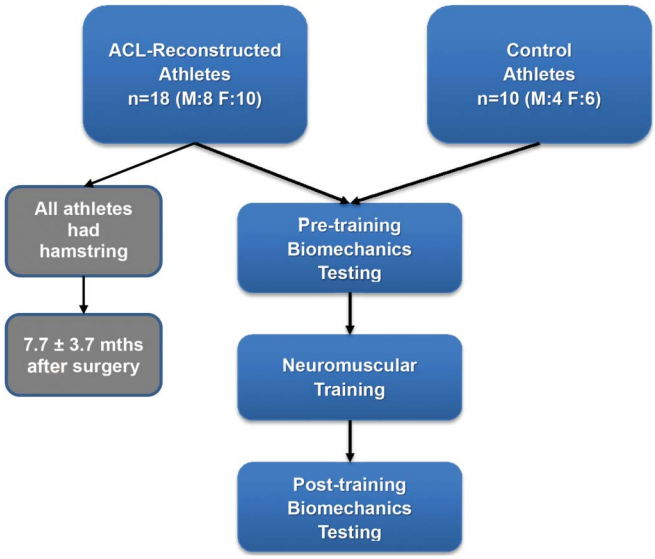

Figure 1. Flowchart of participants and study progression.

RESULTS

Effect of Neuromuscular Training on Knee Biomechanics in Anterior Cruciate Ligament Reconstruction Cohort

There were no significant interactions ($P > 0.05$) between session and limb for knee flexion angle at IC (Figure 2), knee abduction angle at IC (Figure 3), knee flexion moment at IC (Figure 4), and knee abduction moment at IC (Figure 5). A significant main effect of session was observed for knee flexion angle at IC ($P = 0.001$; Figure 2) and knee flexion moment at IC ($P = 0.008$; Figure 4). The ACLR group landed with greater knee flexion angle at IC (pre-training: 17.2 ± 7.1 degrees; post-training: 24.3 degrees ± 9.04 degrees) and lower knee flexion moment at IC (pre-training: 0.48 ± 0.14 Nm/kg; post-training: 0.36 ± 0.10 Nm/kg) after participating in the NMT program. There were no significant main effects ($P > 0.05$) observed for knee abduction angle at IC and knee abduction moment at IC.

There were no significant interactions ($P > 0.05$) for peak knee flexion angle, peak knee abduction angle, peak knee flexion moment, and peak knee abduction moment. A significant main effect of limb was observed for peak knee flexion moment ($P = 0.005$). The uninvolved limbs (1.82 ± 0.1 Nm/kg) demonstrated greater peak knee flexion moment than the involved limbs (1.55 ± 0.01 Nm/kg). Otherwise, there were no significant main effects ($P > 0.05$) or session or limb for peak flexion angle, peak knee abduction angle, and peak knee abduction moment.

There were no significant changes ($P > 0.05$) in LSI from pre-training to post-training for knee flexion angle at IC (pre-training: 107.0 ± 35.7; post-training: 108.0 ± 32.1; $P = 0.47$), peak knee flexion angle (pre-training: 100.1 ± 4.28; post-training: 98.5 ± 3.5; $P = 0.11$), knee flexion moment at IC (pre-training: 98.3 ± 22.0; post-training: 102.8 ± 18.4; $P = 0.28$), and peak knee flexion moment (pre-training: 92.1 ± 39.2; post-training: 85.6 ± 18.5; $P = 0.27$).

Post–Neuromuscular Training Comparison: Athletes After Anterior Cruciate Ligament Reconstruction Versus Controls

There were no significant interactions ($P > 0.05$) between the factors group and limb for knee flexion angle, knee flexion moment, knee abduction angle, and knee abduction moment. A significant main effect of group was observed for knee frontal plane angle ($P = 0.002$). The control group (post-training: 2.33 ± 2.8 degrees) demonstrated a larger knee abduction angle than the ACLR group (post-training: −0.24 ± 2.77 degrees). There was no significant main effect ($P > 0.05$) of group or limb for the other biomechanical variables.

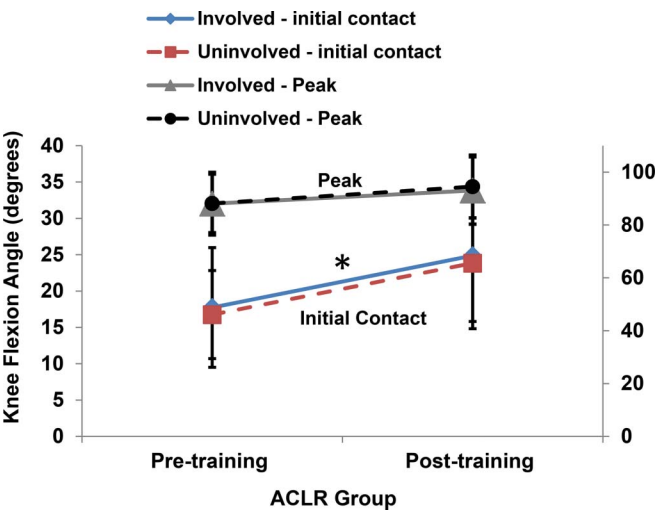

Figure 2. Changes in knee flexion angle at IC from pre-training to post-training for the ACLR group. *$P = 0.001$. Figure 2 can be viewed in full color online at cjsportmed.com.

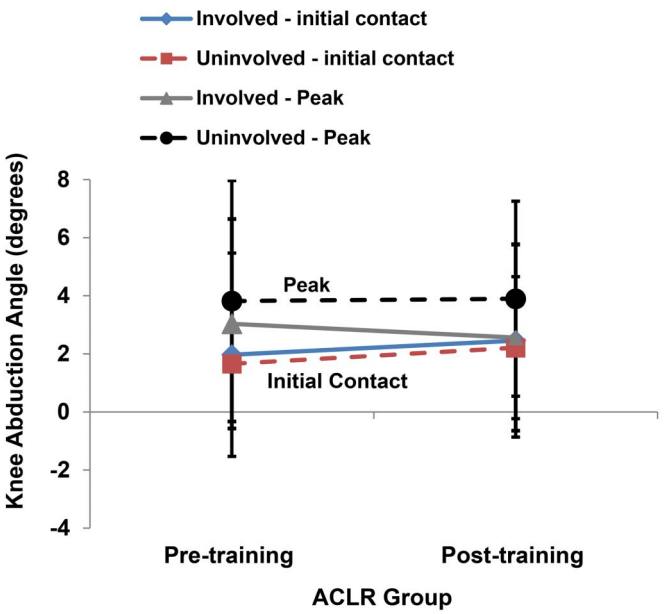

Figure 3. Changes in knee abduction angle at IC from pre-training to post-training for the ACLR group. Figure 3 can be viewed in full color online at cjsportmed.com.

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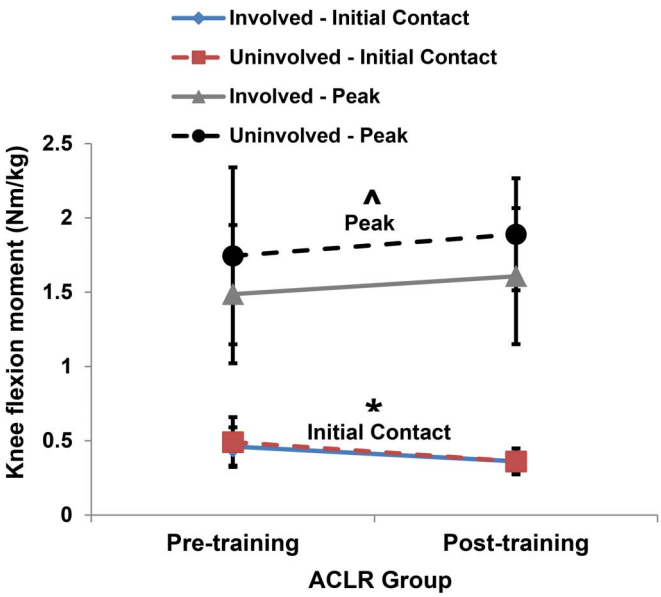

Figure 4. Changes in knee flexion moment at IC from pre-training to post-training for the ACLR group. *P = 0.008. ^P = 0.005. Figure 4 can be viewed in full color online at cjsportmed.com.

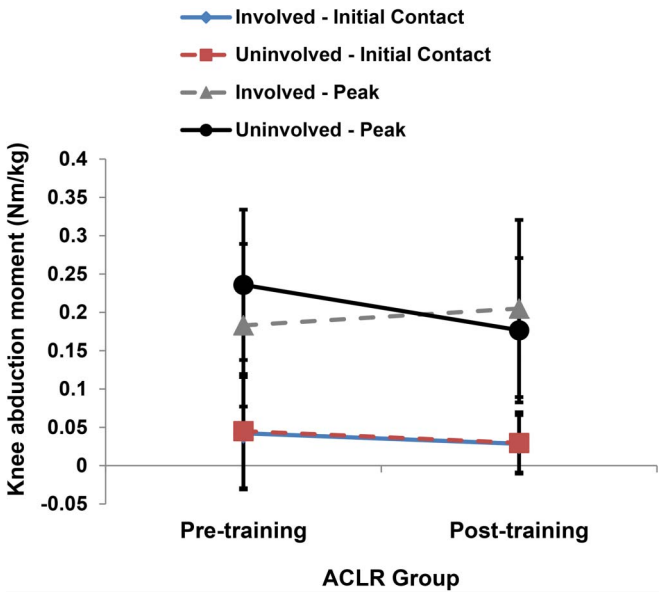

Figure 5. Changes in knee abduction moment at IC from pre-training to post-training for the ACLR group. Figure 5 can be viewed in full color online at cjsportmed.com.

There were no significant interactions or main effects ($P > 0.05$) between the factors group and limb for peak flexion angle, peak abduction angle, peak flexion moment, and peak abduction moment. Table 3 displays a summary of the post-training knee kinematic and kinetic variables for the ACLR and control groups.

DISCUSSION

The athletes with ACLR demonstrated a significant improvement in sagittal plane knee biomechanics after completing the NMT program. The post-training comparison of knee biomechanics between the control and ACLR groups demonstrated comparable knee biomechanics. Participation in such training programs continues to be beneficial for populations vulnerable to lower extremity injury. Previous research on the efficacy of NMT programs has largely focused on uninjured athletes. However, in this study, we demonstrated that these training programs may augment athletes recovering from devastating knee injuries such as an ACL tear before returning to activity. Stiff kinematic and kinetic landing strategies in prospective biomechanical studies have shown to increase the risk of ACL injuries.[34–37] The improvement of athletes with ACLR athletes who landed with greater knee flexion angle and less knee flexion moment after participating in the training program may help mitigate their risk of future ACL injury. The current literature contains limited evidence of the use of an NMT program in ACLR athletes to improve biomechanics and neuromuscular control. Recent studies have investigated the use of perturbation training, a particular type of NMT focused on eliciting coordinated muscle responses to surface perturbations, to resolve impairments in ACL-injured or ACLR athletes. Failla et al[38] compared the functional outcomes 2 years after ACLR in a cohort that underwent preoperative rehabilitation, which included NMT and progressive strengthening with a nonexperimental group. The group that was treated with the NMT and strength training preoperatively had greater functional outcomes as measured by the International Knee Documentation Committee (IKDC) subjective knee form and Knee Injury and Osteoarthritis Outcomes Score (KOOS) and RTS rates 2 years after ACLR.[38] In addition, recent reports from the Anterior Cruciate Ligament–Specialized Post-Operative Return-to-Sports trial, a single-blinded randomized clinical study of secondary prevention and RTS, found similar outcomes for individuals who were either randomized into the strengthening, agility, and secondary prevention (SAP) protocol or the SAP plus additional perturbation training (SAP + PERT).[39,40] Importantly, this secondary injury program is a combination of strengthening and movement training, including plyometric and agility exercises.[39] From these studies, Arundale et al[39] found few differences in men assigned to either SAP or SAP + PERT in measures of quadriceps strength symmetry, single-legged hopping, and patient-reported outcomes. Capin et al[40] found that in the same group of men, the SAP or the SAP + PERT group did not restore gait symmetry 1 or 2 years after ACLR. However, improvements in gait asymmetries were observed in this cohort between 1 and 2 years. The data from our study and the evidence in the literature indicate the potential role of NMT programs to improve self-reported outcomes and landing biomechanics in athletes with ACLR.

Notably, the hypothesis that a significant change in frontal plane kinematics and kinetics would occur in the ACLR group after the NMT program was not supported. This is an unexpected finding because the NMT program was designed to target the frontal plane collapse of the knee during a jump-landing task. Increased knee valgus during a DVJ has been shown to be directly implicated in both primary and secondary ACL injury.[36] The absence of significant measurable changes in the frontal plane after NMT may be due to the

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TABLE 3. Post-training Knee Kinematic and Kinetic Variables of the ACLR and Control Group

Variables (Angles: Degrees Moments: Nm/Kg)	ACLR Group				Control Group				Interactions/ Main Effects and *P*
	Post-training Involved		Post-training Uninvolved		Post-training Involved		Post-training Uninvolved		
Knee flexion angle (IC and peak)	24.9 ± 9.0	93.0 ± 11.9	23.8 ± 9.0	94.5 ± 12.7	22.0 ± 8.0	92.8 ± 9.7	23.5 ± 7.0	93.2 ± 9.8	*Main effect of group: $P = 0.002$
Knee abduction angle (IC and peak)	*2.4 ± 3.3	2.6 ± 3.2	*2.2 ± 2.4	3.9 ± 3.4	*−0.6 ± 2.5	4.4 ± 2.0	*0.17 ± 3.2	5.6 ± 4.0	
Knee flexion moment (IC and peak)	0.36 ± 0.05	1.61 ± 0.5	0.36 ± 0.08	1.89 ± 0.4	0.33 ± 0.08	1.82 ± 0.3	0.31 ± 0.07	1.70 ± 0.4	
Knee abduction moment (IC and peak)	0.03 ± 0.04	0.20 ± 0.1	0.03 ± 0.04	0.18 ± 0.1	0.02 ± 0.02	0.24 ± 0.12	0.003 ± 0.03	0.23 ± 0.1	

There was a significant main effect of group ($P = 0.002$) for knee abduction angle at initial contact. The athletes with ACLR landed with more knee adduction angle at initial contact than the control group.

patient selection process. The study implemented strenuous clinical criteria to ensure participant safety, and we recognize this may have selectively biased our cohort of study to the highest functioning individuals after ACLR. A part of the clinical criteria to participate in the study was that the athlete should demonstrate safe single-leg hopping in place without deleterious landing strategies. Athletes who demonstrated the deleterious medial collapse of the knee during the jumping task were excluded due to its association with increased risk of second ACL injury. Therefore, this ACLR cohort included athletes who already demonstrated frontal plane control of the knee. In addition, the ACLR athletes completed post-operative rehabilitation within the institution's physical therapy department, which practices evidence-based rehabilitation guidelines and addresses neuromuscular control of the knee within the frontal plane. These factors may have confounded the effect of training on knee frontal plane biomechanics. Regardless, the ACLR athletes still demonstrated an improvement in knee biomechanics after completing an NMT program before returning to sport.

There are a few limitations to the current study. There was a limited size within the control group. However, the literature has already shown that NMT is effective in modifying landing biomechanics in uninjured athletes. The aim was to demonstrate that the ACLR athletes after participating in the NMT are comparable with a group of control athletes who completed the same training program. Again, the strict clinical criteria to participate in the study may have restricted us from selecting some athletes who would have greatly benefited from the NMT. However, our first priority was the safety of the athletes and to perform some of the dynamic movements involved in the training the athletes were required to demonstrate some baseline health. The only requirement for the athletes participating in the study was to complete 12 training sessions. Therefore, the athletes did not complete all exercises within the training program. The decision to advance the athlete was made by the trainer and was based on their judgment of the athlete demonstrating proper technique of the exercise for at least 3 sets. Because of this performance-based advancement, not all the athletes were at the same performance level at the end of the training program. Regardless, these athletes demonstrated improvement in landing biomechanics. In addition, another control that will be considered in future studies is an ACLR group without any training because it is important to consider that individuals after ACLR might demonstrate improvements over time.

CONCLUSIONS

The purposes of this study were to: (1) investigate the effects of an NMT program on knee biomechanics in a cohort of ACLR athletes before returning to sport; and (2) compare the post-training knee biomechanics of the ACLR cohort to a control cohort who also completed the same NMT program. The high-functioning ACLR group demonstrated an improvement in sagittal plane knee biomechanics and neuromuscular control after participation in the NMT program. Furthermore, after the training, the group of ACLR athletes demonstrated similar knee biomechanics and neuromuscular control to a group of uninjured control athletes. Movement impairments may be the last of the residual deficits to normalize after ACLR, and some athletes may never regain normal knee function.[41] Recently, evidence in the literature has highlighted that those athletes who wait to return to preinjury level of activity until at least 9 months after ACLR may mitigate their risk of second ACL injury.[41,42] A structured NMT program that exclusively focuses on improving biomechanics and neuromuscular control may remediate these aberrant movement patterns. Future work will be required to understand if NMT in the ACLR population mitigates the risk of a second ACL injury rates and how it compares against strength training and home-based exercise programs.

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